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CGSCNet: ConvGRU Based Skip-connection Framework for Hyperspectral Image Super-Resolution

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Abstract

Hyperspectral image super-resolution has been extensively studied, and significant development has been made based on deep convolutional neural networks. However, in most residual super-resolution networks, straightforward skip-connections spatial features with different levels of abstraction can lead to semantic confusion, ultimately affecting reconstruction accuracy. To address this issue, we developed a ConvGRU based skip-connection strategy. This strategy leverages ConvGRU to filter features across consecutive layers, enhancing feature propagation and fusion. Accordingly, we develop a ConvGRU based skip-connection framework designed specifically for hyperspectral image super-resolution (CGSCNet). The framework consists of three parts. The first module is spatial feature reconstruction, which introduces ConvGRU to handle the fusion of semantic information across different layers and gradually builds spatial-spectral information by leveraging the feature reuse properties of the residual network. The second module is detail refinement, aimed at enhancing spatial detail information. The third module, spectral information reconstruction, recalibrates spectral responses by modeling the relationships between different spectral bands. Experiments on the Chikusei, and Houston datasets demonstrate that our framework outperforms state-of-the-art comparison models in both quantitative evaluation metrics and visual quality across a wide range of scenarios.

CCS Concepts

• Computing methodologies; • Artificial intelligence; • Computer vision; • Computer vision tasks;

*Peng Ren is the corresponding author.

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Keywords

ConvGRU, detail refinement, feature fusion, hyperspectral image, super-resolution

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1 INTRODUCTION

Hyperspectral images (HSI) typically include dozens to hundreds of bands [1], offering extensive spectral information that accurately characterizes the intrinsic structure and material properties of objects [2]. Consequently, HSI is widely used in various applications such as geological exploration [3], land cover classification [4], and change detection [5]. Recently, application tasks have shifted from scene-level recognition to pixel-level recognition, significantly raising the accuracy requirements. Consequently, remote sensing satellites must capture images with higher spatial resolution and more detailed spectral information. However, due to limitations in physical devices and imaging time constraints, remote sensing satellites cannot capture hyperspectral images with high spatial resolution. Thus, improving the spatial resolution of HSI is highly significant. Hyperspectral image super-resolution (SR) is an effective technique for enhancing image quality by converting low spatial resolution images into high spatial resolution images through post-processing, without modifying the physical equipment.

1.1 Limitations of deep learning based HSI super-resolution

1) Indiscriminate the fusion of semantic information across different layers: Features from different layers represent varying levels of abstraction in hyperspectral super-resolution. However, many deep learning methods, particularly residual networks, use skip-connections too simplistically, fusing features without careful consideration.

Wang et al. [6] proposed the group-based embedding learning and integration network (GELIN), which uses a neighboring group ensemble approach to reduce the complexity of feature extraction and reconstruction. Jiang et al. [7] proposed a spatial-spectral prior

for super-resolution (SSPSR), which uses a spatial-spectral feature extraction block with efficient residual learning and attention mechanisms, fully exploiting spatial-spectral prior information. Li et al. [8] proposed a dual-stage super-resolution (DualSR) method, in which the coarse stage performs band-wise super-resolution with an alternating spectral fusion mechanism, while the fine stage applies an enhanced back-projection method to learn spatial-spectral consistency, thus improving reconstruction performance.

These methods use skip-connections to transmit and fuse features across layers, expanding the receptive field and enhancing representation. However, passing all low-level features indiscriminately can cause semantic inconsistencies and redundancy, reducing model performance.

2) Preserving Spatial and Spectral Precision: Hyperspectral images capture a wide variety of ground scenes, but their spatial and spectral information is frequently compromised by excessive erosion and complex degradation processes. Consequently, recovering fine details and sharp edges from low-resolution inputs is significantly more challenging than from natural images.

Hui et al. [9] introduced a model for predicting detail components to enhance image details and recover fine-grained information. Pan et al. [10] presented a texture-enhanced ResUNet model, incorporating a texture enhancement module to bolster the texture details in lower-level features. Zhang et al. [11] developed an edge-enhanced efficient network utilizing edge-enhanced convolutional blocks and multi-step gradient extraction techniques, among other feature enhancement methods, to strengthen edge perception.

These methods assume clear input image details, enhancing details based on this. However, current deep learning-based hyperspectral super-resolution struggles with excessive spatial and spectral erosion, leading to blurred textures and noise, making detail recovery challenging.

1.2 Contributions of our work

Our novel contributions to hyperspectral image super-resolution are presented as follows.

1) To prevent indiscriminate feature transmission and fusion, we developed a ConvGRU skip-connection unit. This method employs gating and memory balancing mechanisms to intelligently control feature retention and updates, facilitating effective feature transmission and fusion across layers. As a result, it facilitates precise and effective transmission and fusion of features with varying semantics.

2) To preserve spatial and spectral accuracy, we established a hyperspectral image super-resolution framework comprising three components. Each component is based on ConvGRU skip-connections and is dedicated to spatial feature reconstruction, detail refinement, and spectral information reconstruction. This framework maintains both spatial and spectral accuracy in hyperspectral image super-resolution.

3) Experimental results on the Chikusei and Houston datasets show that the proposed framework significantly outperforms other methods in hyperspectral image super-resolution. The code for the proposed framework is publicly available to facilitate further evaluation.

2 CONVGRU SKIP-CONNECTION

In order to demonstrate the progression from the straightforward skip-connection to our novel ConvGRU skip-connection, both types of skip-connections are illustrated in Figure 1. In neural network structures, skip-connections can create short paths from lower-level to higher-level features, which is considered an effective method for alleviating the challenges of training deep neural networks. Since features from different depths and receptive fields exhibit distinct patterns in skip-connections, previous direct skip-connection approaches may result in mixed feature information during fusion.

Therefore, we designed a ConvGRU-based skip-connection method to regulate the fusion of semantic information with different levels of abstraction. The ConvGRU skip-connection unit is illustrated in Figure 1(b). By comparing the two types of skip-connections, we observed that a straightforward skip-connection simply fuses the feature information from the previous layer into the current layer. In contrast, the ConvGRU skip-connection, based on the principle of gating mechanisms, selectively chooses important feature information from the previous layer. This effectively avoids information redundancy and confusion during the feature fusion process.

In the ConvGRU skip-connection unit, input features bypass additional processing because they already include feature representations from the preceding ConvGRU block. Incorporating these features through addition may introduce redundancy and reduce the ConvGRU's memory function. Additionally, input features in the skip-connection are not processed by activation layers to preserve more raw information for the ConvGRU's update and reset gates. The ConvGRU's update and reset gates use nonlinear activation functions to selectively integrate input information.

3 PROPOSED FRAMEWORK

3.1 Overall Framework

As illustrated in Figure 2, the CGSCNet framework consists of three main components: spatial feature reconstruction, detail refinement, and spectral information reconstruction. First, the spatial feature reconstruction module's primary feature extraction unit captures low-level features from the input low-resolution hyperspectral image. These features are then passed to the deep feature exploration module, which progressively extracts more complex features. This deep feature extraction process incorporates ConvGRU skip-connection units. Next, the detail refinement module learns detailed mappings from the intermediate reconstruction outputs. Finally, the spectral information reconstruction module refines the spectral responses using attention values obtained from the ConvGRU based spectral attention mechanism, generating a final HSI with enhanced details. The reconstructed hyperspectral image preserves rich spatial and spectral information.

3.2 Spatial Feature Reconstruction

Due to the high dimensionality and large volume of hyperspectral data, we implemented a progressive spatial reconstruction strategy to overcome the challenges posed by dimensionality. This strategy consists of three stages: primary feature extraction, deep feature exploration, and upsampling and reconstruction. First, during the

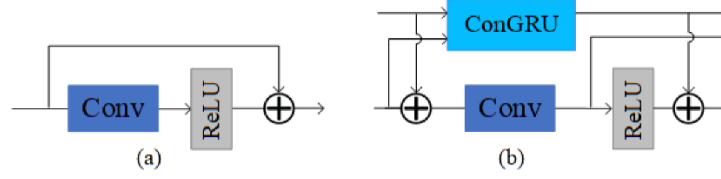


Figure 1: Progress from (a) straightforward skip-connection to (b) our novel ConvGRU skip-connection.

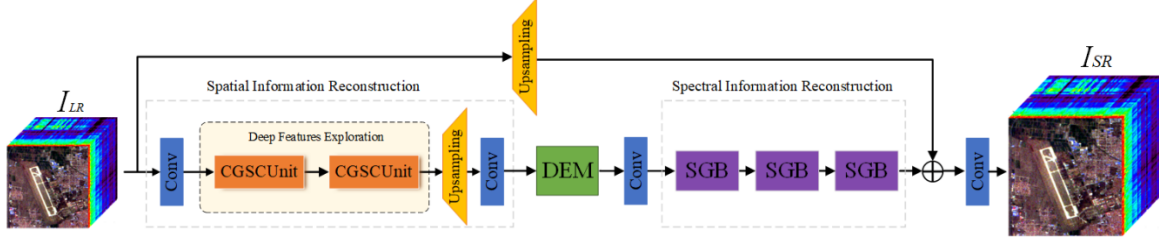


Figure 2: Proposed ConvGRU skip-connection based hyperspectral image super-resolution framework.

primary feature extraction stage, the input image undergoes initial feature extraction using a 3×3 convolutional layer:

$$F_P = \text{Hpfe}(I_{LR}) \quad (1)$$

where I_{LR} denotes the low-resolution input and F_P represents the low-level features extracted from this input.

In the deep feature exploration stage, we employ the ConvGRU based skip-connection unit (CGSCUnit) to standardize feature fusion across different abstraction levels. Let N denote the number of CGSCUnits stacked in the stage. Let F_{N-1} represent the input features of the N -th CGSCUnit and F_N denote its output features. When $N=1$, $F_{N-1}=F_P$. This process is expressed as follows:

$$F_N = \text{CGSCUnit}_N(F_{N-1}) \quad (2)$$

Finally, our framework employs a staged upsampling approach that progressively enhances image details through iterative phases. By breaking down the super-resolution reconstruction process into multiple steps, the framework reduces the challenges and complexity associated with high-dimensional data. The structure of the upsampling and reconstruction network is outlined below:

$$F_{SR}^k = \text{Conv}_k(\text{Upsampling}_k(F_N)) \quad (3)$$

where k denotes the k -th upsampling and reconstruction stage, $\text{Upsampling}(\cdot)$ refers to the sub-pixel convolution upsampling, and $\text{Conv}(\cdot)$ represents the convolution operator. The image F_{SR}^k denotes the intermediate reconstruction result obtained from upsampling. Convolutional layers are added immediately after the upsampling layers to help reduce artifacts in the super-resolution output.

3.3 Detail Refinement

The spatial information in hyperspectral images is often overly eroded, leading to blurry details and significant noise when directly extracting features. To address this, we developed a detail refinement model to extract detailed features from the intermediate results F_{SR}^k produced by the spatial information reconstruction module. The Detail Refinement model consists of wavelet

transform-based detail modulation blocks, as illustrated in Figure 3.

Our wavelet transform-based detail modulation block efficiently extracts and enhances detail information in reconstructed hyperspectral images. This module models multi-scale features in the frequency domain, allowing the extraction and enhancement of both global and local details. Specifically, the feature modulation process begins with the input feature map F_{SR}^k , the process described into the following:

$$F_{SR} = \text{Concat} \left(\begin{array}{l} \text{Conv} \left(\text{Wavelt} \left(\text{Conv} \left(F_{SR}^k \right) \right) \right) \otimes \text{Conv}_7 \left(F_{SR}^k \right), \\ \text{Wavlt} \left(\text{Conv} \left(F_{SR}^k \right) \right), \text{Conv}_7 \left(F_{SR}^k \right) \end{array} \right) \quad (4)$$

where $\text{Conv}(\cdot)$ represents a convolution used to extract fine details, whereas $\text{Conv}_7(\cdot)$ is a 7×7 convolution utilized for preserving broader details. The $\text{Wavelt}(\cdot)$ operator denotes wavelet transform modulation, which produces four directional components: cA (approximation), cH (horizontal detail), cV (vertical detail), and cD (diagonal detail). This modulation adaptively enhances pertinent details while suppressing noise. The final output, F_{SR} , is the fused feature map that combines both local and global details.

3.4 Spectral Information Reconstruction

Spectral dimension information is a critical component of hyperspectral images. While spatial feature reconstruction and detail refinement primarily address spatial information, they may overlook spectral dimension details. To address this gap, we propose a spectral information reconstruction module. This module comprises M spectral guided blocks, as shown in Figure 4. The spectral guided block is designed to model interactions between different spectral bands comprehensively. It includes CGSCUnits and spectral attention mechanism. The CGSCUnits extract spectral information, while the Spectral Attention mechanism, similar to channel attention, adjusts the spectral response by learning attention weights

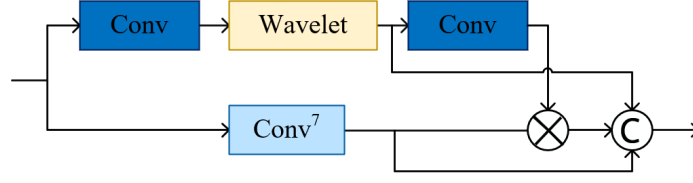


Figure 3: The wavelet transform-based detail modulation block within the detail refinement model.

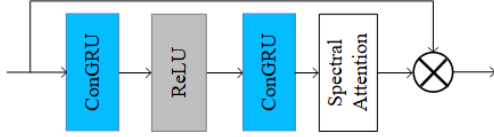


Figure 4: ConvGRU skip-connection based spectral information reconstruction with spectral guided block

during the channel stretching process. This approach helps maintain the accuracy of spectral information.

Specifically, the Spectral Attention mechanism recalibrates the weights of each channel by computing matrix coefficients. The full-band features $F_q \in \mathbb{R}^{H \times W \times C}$, where H and W represent the spatial dimensions and C is the number of feature channels. Three linear layers are used to learn the matrix coefficient relationships between different spectral bands, resulting in matrix coefficient attention values W .

Gradually increasing the number of channels allows for capturing low-level spectral features while improving the learning of high-level feature representations. Conversely, abruptly reducing the number of channels helps the model focus on selecting the most important feature channels. The Spectral Guided Block, which guides spectral information recalibration, is represented as $\text{SGB}(\cdot)$, operating in the form of $\text{wConvGRU}(\text{ConvGRU}(\cdot))$. Therefore, spectral information reconstruction is as follows:

$$F_q = \text{SGB}_M (\text{SGB}_{M-1} (\dots (\text{SGB}_1 (F_q)) \dots)). \quad (5)$$

Next, F_q is added element-wise to the upsampled input image. Finally, a convolution layer is applied to produce the final reconstructed image I_{SR} :

$$I_{\text{SR}} = \text{Conv} (\text{Add} (\text{Conv} (F_q), \text{Upsampling} (I_{\text{LR}}))) \quad (6)$$

4 EXPERIMENTAL RESULTS AND ANALYSIS

4.1 Dataset and Evaluation Metrics

Our framework was evaluated using real data from diverse urban and rural scenarios. The Chikusei dataset includes both agricultural and urban areas in Japan. It contains 128 bands (363–1,018 nm), with a spatial resolution of 2.5 meters, and covers an area of $2,304 \times 2,048$ pixels. The Houston dataset captures hyperspectral data from the University of Houston campus. It consists of 48 bands (380–1,050 nm) with a 1-meter resolution and dimensions of $4,172 \times 1,202$ pixels. Four key evaluation metrics—Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), Spectral Angle Mapper (SAM), and Correlation Coefficient (CC)—were used to assess spatial, perceptual, and spectral performance.

4.2 Implementation Details

The datasets are divided into 80% for training and 20% for testing, with scenes segmented into two distinct regions. For a scaling factor of 4, data is processed into patches with a 32-pixel overlap. The degradation model uses Disk blur kernels and bicubic interpolation to produce low-resolution images, resulting in 16×16 pixel outputs. The framework is implemented in PyTorch and executed on an NVIDIA GeForce RTX A6000 GPU paired with an AMD EPYC 7742 CPU. The CGSCUnit N is set to 12, the spectral guided block P is set to 3, and 96 low-resolution images are randomly chosen for each training batch.

4.3 Performance Evaluation

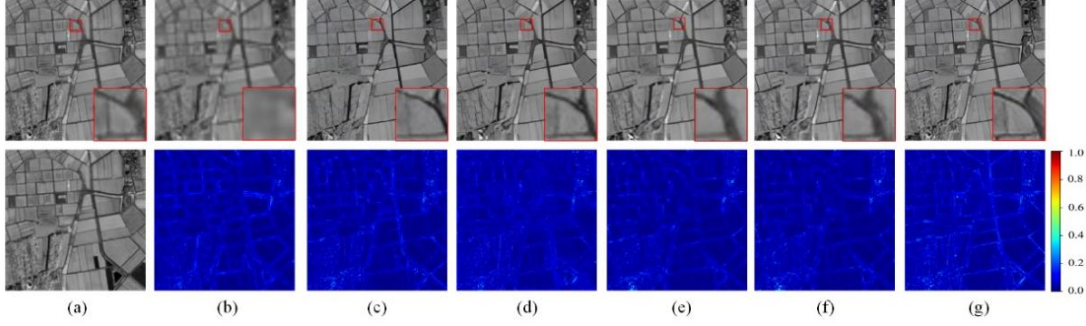
The CGSCNet is compared to five leading hyperspectral image super-resolution methods with a $\times 4$ upsampling factor: Bicubic interpolation, SSPSR, DualSR, GELIN, and EUNet [12]. Bicubic is a standard resizing method, while SSPSR uses advanced residual blocks. DualSR focuses on grouping similar spectral bands, GELIN applies group-based embedding learning, and EUNet integrates explicit constraints from the degradation model. Publicly available implementations of these methods in PyTorch are used, following the original parameter settings and applied to the same datasets.

1) Experimental Results on Chikusei Dataset: **Quantitative Results:** Table I summarizes the average reconstruction performance of CGSCNet and five other methods on the Chikusei dataset, with the best outcomes highlighted in bold and the second-best underlined. CGSCNet surpasses all other methods in PSNR, SSIM, and SAM, showing a 0.3834 dB improvement in PSNR compared to the state-of-the-art approaches. The Bicubic method delivers sub-optimal results, while models such as GELIN and EUNet achieve better performance by integrating spatial and spectral information. CGSCNet excels by effectively merging semantic information at different abstraction levels, refining details through the Detail Refinement module, and preserving spectral correlations using the Spectral Guided Block.

Visual Results: Figure 5 shows the spatial super-resolution results for a 93-band image from various test areas, enabling qualitative comparison across methods. To enhance clarity, regions marked by red rectangles are enlarged. Compared to other methods, CGSCNet offers the most accurate and realistic visual effects, particularly in restoring edge details. Other methods display blurry or distorted reconstructions. The Bicubic method cannot add fine details, and while deep learning methods like DualSR, GELIN, and EUNet reconstruct some textures, they struggle to fully capture shallow edge contours due to limitations in global optimization and insufficient spatial feature utilization. This is particularly evident in the left-hand target area within the red rectangles.

Table 1: Quantitative results of hyperspectral image super-resolution on the Chikusei dataset.

Metric	Bicubic	SSPSR [7]	DualSR [8]	GELIN [6]	EUNet [12]	CGSCNet
PSNR↑	35.5357	40.2279	39.8686	40.1423	40.2534	40.6368
SSIM ↑	0.7235	0.9424	0.9292	0.9371	0.9429	0.9451
SAM ↓	3.404	2.4575	3.1043	2.4514	2.4495	2.4335
CC ↑	0.9074	0.9555	0.9448	0.9474	0.9568	0.9551

**Figure 5: Quantitative results of hyperspectral image super-resolution on the Chikusei dataset. (a) Ground truth, (b) Bicubic, (c) SSPSR [7], (d) DualSR [8], (e) GELIN [6], (f) EUNet [12], (g) CGSCNet.****Table 2: Quantitative results of hyperspectral image super-resolution on the Houston dataset.**

Metric	Bicubic	SSPSR [7]	DualSR [8]	GELIN [6]	EUNet [12]	CGSCNet
PSNR↑	32.2735	36.2978	36.2602	36.2364	36.3111	36.6133
SSIM ↑	0.7262	0.9376	0.9368	0.9355	0.9386	0.9390
SAM ↓	2.6523	2.1812	2.0224	2.0247	2.0209	2.0007
CC ↑	0.9561	0.9766	0.9764	0.9763	0.9768	0.9788

Figure 5 presents the absolute error maps for various methods for a 93-band image. These maps highlight the discrepancies between the reconstructed images and the ground truth, with smaller error areas indicating better performance of the super-resolution methods. In Figures 5, CGSCNet exhibits notably lower errors. This result underscores that CGSCNet achieves superior spatial accuracy in reconstructing details, particularly in restoring edges and textures in fields. This can be attributed to CGSCNet’s Detail Refinement model, which reduces edge noise interference and enhances spatial fidelity. Figure 7a shows the spectral curves of selected pixels from the Chikusei dataset, which closely align with the ground truth in most cases. This highlights CGSCNet’s strength in preserving spectral information, demonstrating the effectiveness of the Spectral Information Reconstruction module.

2) Experimental Results on Houston Dataset: **Quantitative Results:** Table 2 shows the average performance of CGSCNet and five other methods on the Houston dataset. CGSCNet achieves PSNR of 36.6133 dB, SSIM of 0.9390, SAM of 2.0007, and CC of 0.9788, improving PSNR by 0.2715 dB over other methods. **Visual Effects:** Figure 6 compares reconstruction results for the 39th band, with key regions highlighted in red. Bicubic interpolation blurs edges, and SSPSR, DualSR, and GELIN introduce artifacts, while EUNet oversharpen. CGSCNet offers the most accurate edge and contour

details, closely matching the ground truth. Figure 6 shows error maps for the 39th band, where CGSCNet excels in reconstructing high-frequency edges and low-frequency textures. Figure 7b shows spectral curves from selected pixels, highlighting CGSCNet’s superior spectral fidelity over other methods.

5 CONCLUSION

This article presents a novel hyperspectral image super-resolution framework using a ConvGRU-based skip-connection to model continuous layer features. Unlike traditional methods, this approach selectively filters and integrates key features across layers, enhancing feature fusion. The framework consists of three components: ConvGRU based spatial feature reconstruction, detail refinement, and spectral information reconstruction. The ConvGRU skip-connection improves spatial data transfer, while the detail refinement module enhances intricate multi-scale features. The spectral reconstruction component adjusts responses between spectral bands for high fidelity. Our framework outperforms five other approaches in both visual quality and quantitative metrics.

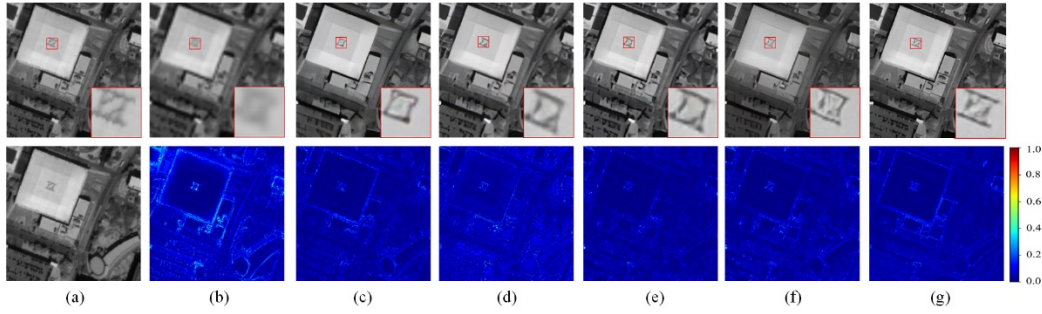


Figure 6: Quantitative results of hyperspectral image super-resolution on the Houston dataset. (a) Ground truth, (b) Bicubic, (c) SSISR [7], (d) DualSR [8], (e) GELIN [6], (f) EUNet [12], (g) CGSCNet.

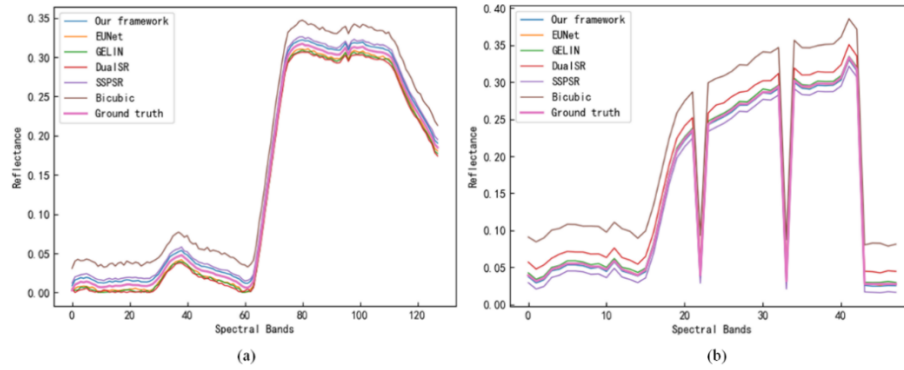


Figure 7: Spectral curves of selected locations on (a) Chikusei, (b) Houston datasets.

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